

Replace the root volume for an Amazon EC2 instance without stopping it

To begin with the lab

Summary of the Lab

To create and work with a Windows Server 2022 EC2 instance, launch a new instance on AWS, set up a key pair and security group, and connect via RDP using the decrypted password. Inside the VM, create a new folder and a text file in the C: drive. Then, go to AWS Console and create a snapshot from the volume. After deleting the folder from the VM, replace the root volume using the snapshot. AWS will stop the instance, replace the root volume, and restart it. Reconnect to the VM to verify that the folder has been successfully restored.

- Let's create an instance first.
- Go to AWS Console.
- Navigate to EC2.
- Click on launch an instance.
- Name you instance.
- In Application and OS Images (Amazon Machine Image), go to quick start
- Select Windows (Microsoft)
- In Amazon Machine Image (AMI), Select Microsoft Windows Server 2022 Base.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below.

Recents | My AMIs | **Quick Start**

Amazon Linux | macOS | Ubuntu | **Windows** | Red Hat | SUSE Linux | Debian

Amazon Machine Image (AMI)

Microsoft Windows Server 2022 Base [ami-0545f44fe05216fc4](#) [Free tier eligible](#)

Virtualization: hvm | ENA enabled: true | Root device type: ebs

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Microsoft Windows Server 2022 ...[read more](#)

ami-0545f44fe05216fc4

Virtual server type (instance type)

t2.micro

Firewall (security group)

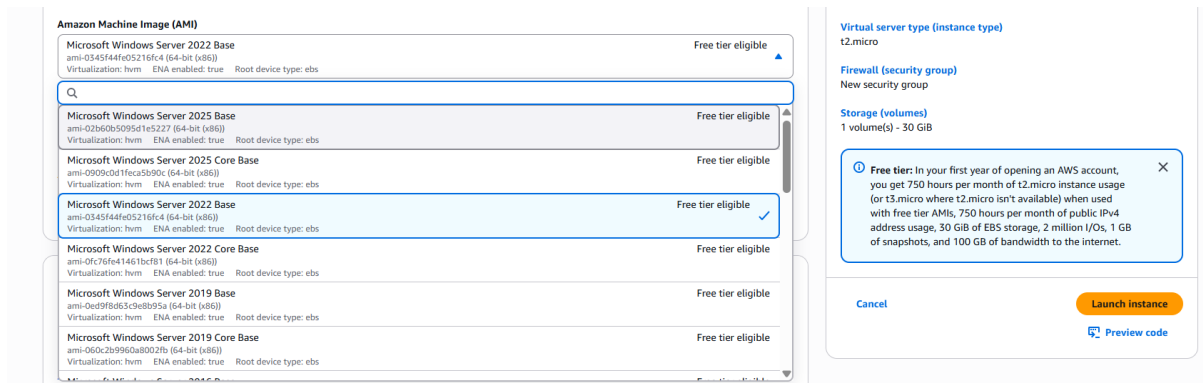
New security group

Storage (volumes)

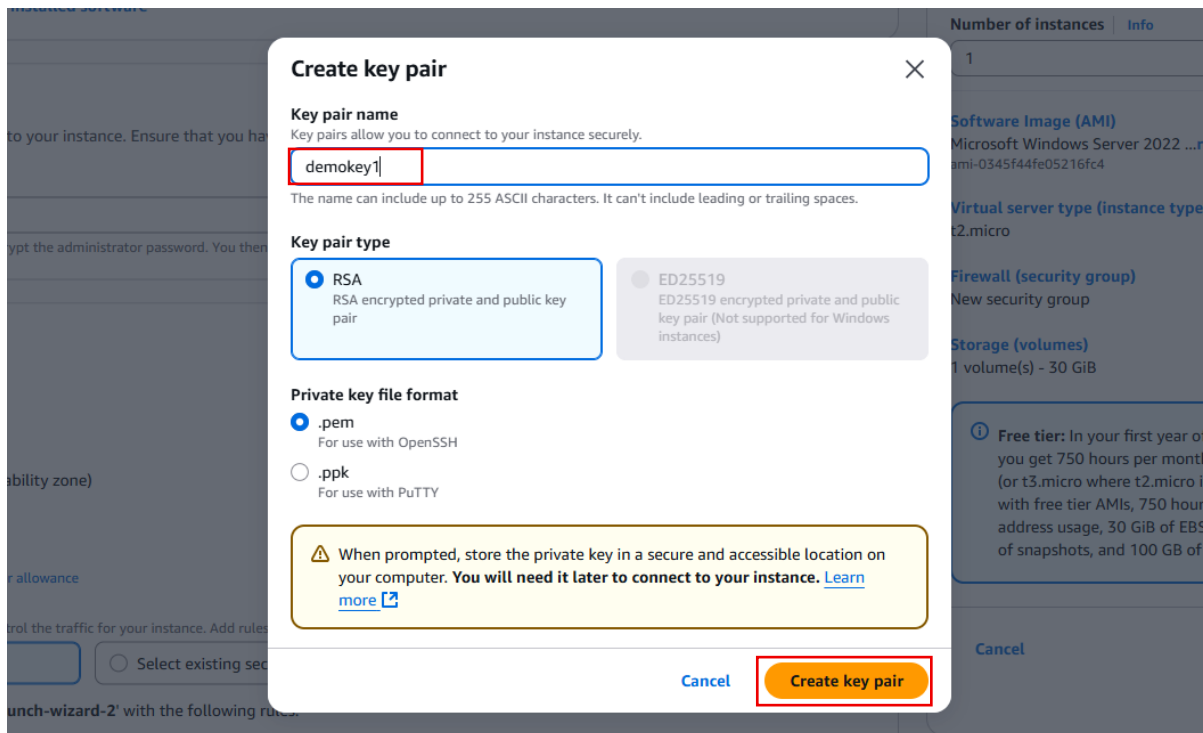
1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Launch instance](#) [Preview code](#)



- Now in key pair (login)
- If you have existing key pair, you can use that.
- Otherwise, click on the ‘Create new key pair’.
- Name your key pair.
- Click on the ‘create key pair’
- Choose the options as given.



- It will download a ‘.pem’ file that will used to login to the virtual machine.
- Go to Network settings
- If you have existing security group, select that.
- Otherwise, click on the create security group
- Choose the option given below.

Network settings [info](#) [Edit](#)

Network [info](#)
vpc-0a79afbc23dd26f1

Subnet [info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [info](#)
Enable
Additional charges apply when outside of free tier allowance

Firewall (security groups) [info](#)
Security groups in place of firewalls help you control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

- ☒ Allow RDP traffic from Helps you connect to your instance Anywhere 0.0.0.0/0
- ☐ Allow HTTPS traffic from the internet To set up an endpoint, for example when creating a web server
- ☐ Allow HTTP traffic from the internet To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Software Image (AMI)
Microsoft Windows Server 2022 ...[read more](#)
ami-0345f44fe05216fc4

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
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[Cancel](#) [Launch instance](#) [Preview code](#)

- For now, I am not increasing the memory size for the root volume.
- You can increase according to your need.
- Click on the launch instance.

Configure storage [info](#) [Advanced](#)

1x 30 GiB gp2 Root volume, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

[Click refresh to view backup information](#) The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

Virtual server type (instance type)
t2.micro

Firewall (security group)
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[Cancel](#) [Launch instance](#) [Preview code](#)

- Here is the new created instance.
- Select the instance.
- Click on the connect.

Instances [info](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Last updated less than a minute ago

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
myvm-server	i-0ed89c2f2ab0b36b1	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-3-85-90-126.comp...	3.85.90.126	-

- Now in connect, go to RDP client.
- Either download the remote desktop file.
- Click on the 'get password'.

Connect [Info](#)
Connect to an instance using the browser-based client.

Session Manager **RDP client** EC2 serial console

[Record RDP connections](#)
You can now record RDP connections using AWS Systems Manager just-in-time node access. [Learn more](#) Try for free ×

Instance ID
i-Oed89c2f2ab0b36b1 (myvm-server)

Connection Type
☒ Connect using RDP client
 Download a file to use with your RDP client and retrieve your password.
 ☐ Connect using Fleet Manager
 To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#).

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS
ec2-3-83-90-126.compute-1.amazonaws.com

Username [Info](#)
Administrator

Password [Get password](#)

[If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.](#)

Cancel

- In the 'get window password'
- Upload the private key file. (upload the key(.pem) that was downloaded.
- Click on the 'Decrypt password'

Get Windows password [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
i-Oed89c2f2ab0b36b1 (myvm-server)

Key pair associated with this instance
demo-key

Private key
Either upload your private key file or copy and paste its contents into the field below.

[Upload private key file](#)

☒ demo-key.pem
1.674KB

Private key contents - optional

```
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEAd8VSHx1dy8kqOYug+FREyk1NUlUX/wdcJSFsVQhvh1G2
WlwV4Xs/3KvVfMoi94oC2FG/ak8ZNCtmYCTKyCZVHMMd7o5pdVQta51KSXFU
FoPBPzpISCC+SOBGA/XmY11jg8/70kS4B3+OGNAmeMc7R8Po1PZmk8NzBjk
YA9td0KvQemHKYvdwKkh+ZMqUL4Ph6v2XV/gBkGae9cCL0W4AcEhuKso9W0lo
E0mRoMC4bsj7j0fX0Rwr9vk3W9NFUdPGAMNdU+ajih3hEN+o4FYW5XID8Ke/3GP
eGb8estLWPS9JAG5dq15LAMycheJaaOgHT4DQQDAQBAoIBAFhtYaZLMVOXOO6v
g3G4KHTApYkXZ2b93lo7VaPx3UtpwQYFpWRelZDYupuj/AhpoYBF1P83Jyn03
-----
```

Cancel [Decrypt password](#)

You will get you password.

Connect [Info](#)
Connect to an instance using the browser-based client.

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When prompted, connect to your instance using the following username and password:

Public DNS
ec2-3-83-90-126.compute-1.amazonaws.com

Username [Info](#)
Administrator

Password
5k%YV [REDACTED]

[If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.](#)

Cancel

- Now copy the Username, password and the public IPv4 address.

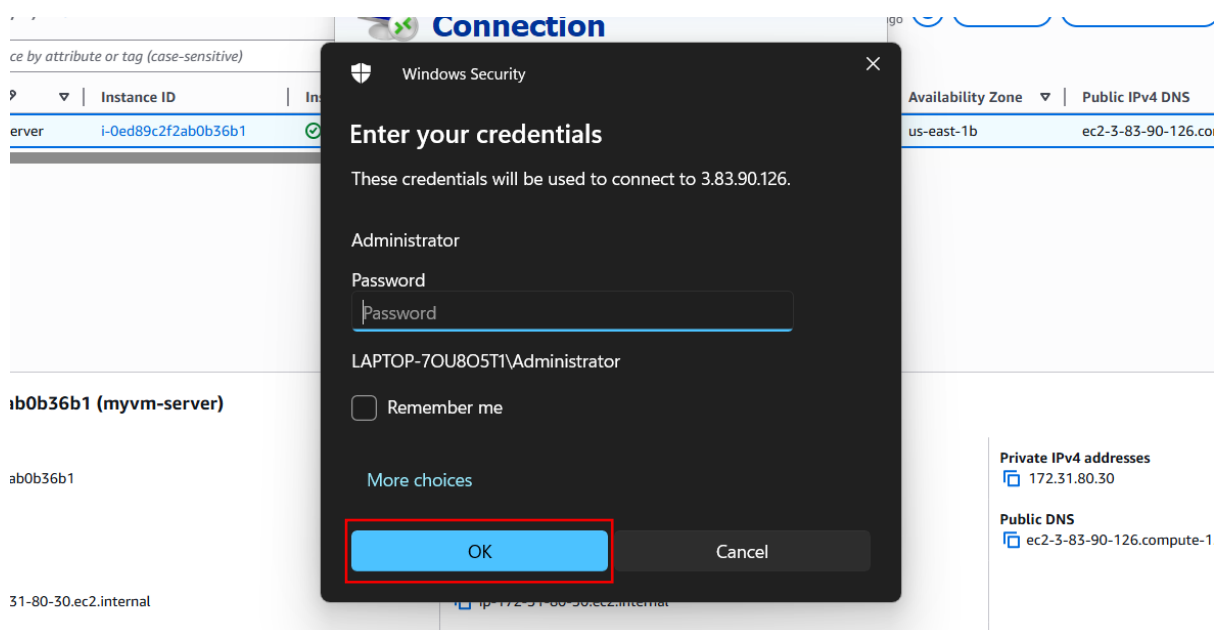
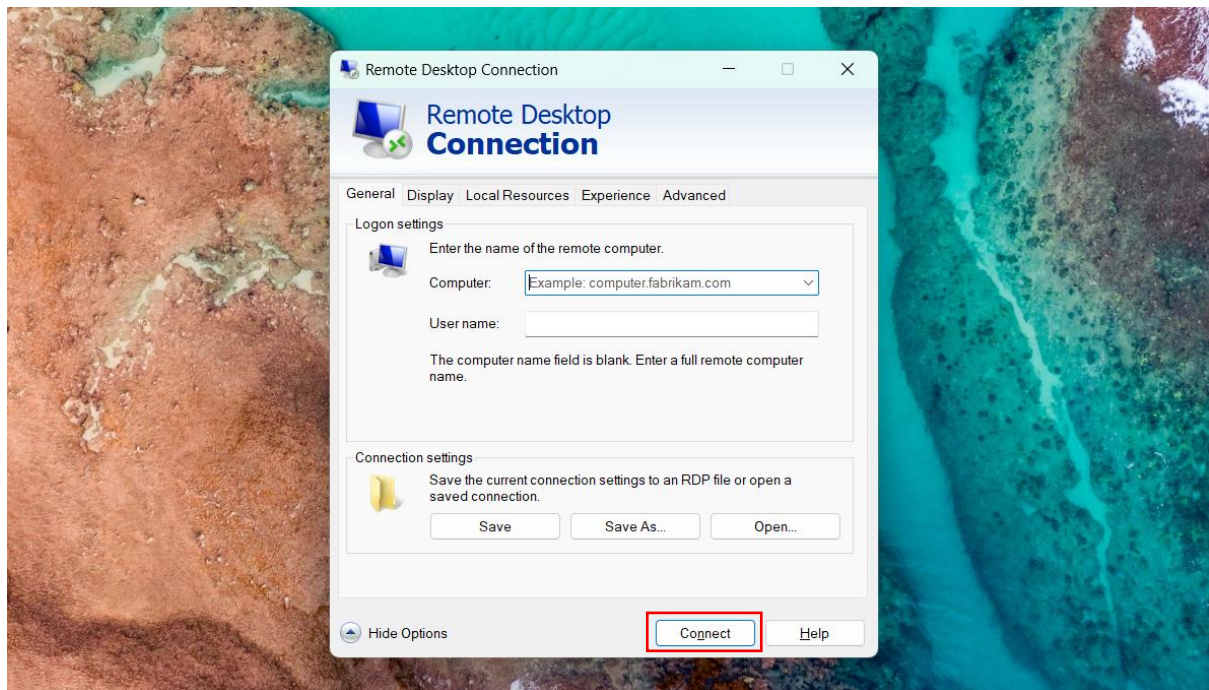
i-0ed89c2f2ab0b36b1 (myvm-server)

Instance ID
i-0ed89c2f2ab0b36b1

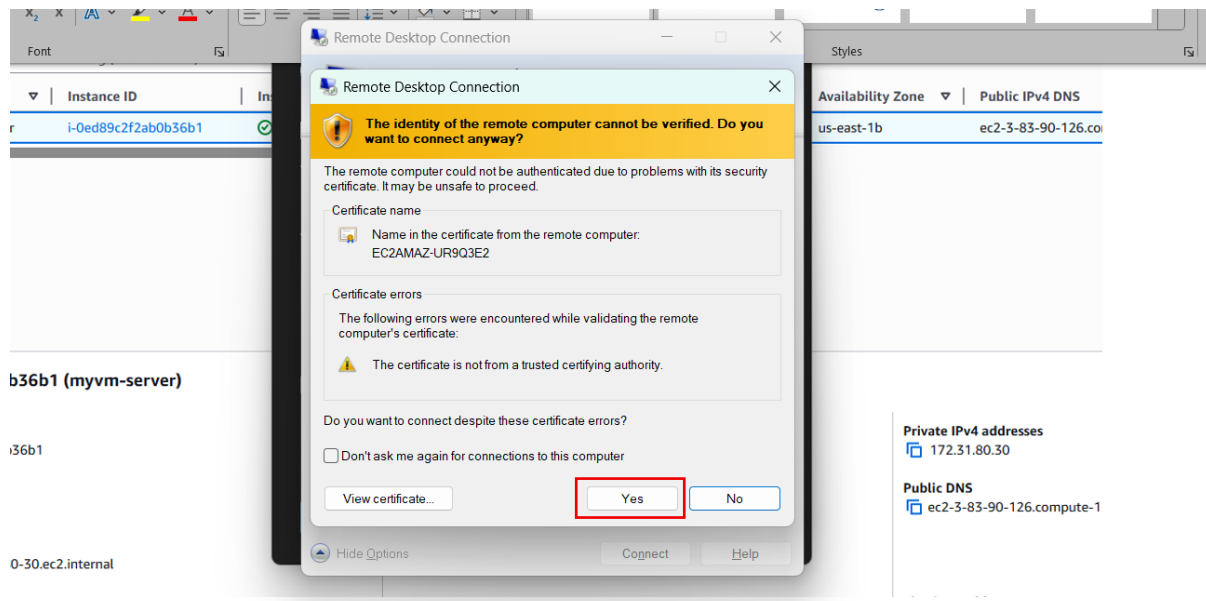
Public IPv4 address
3.83.90.126 | open address

Private IPv4 addresses
172.31.80.30

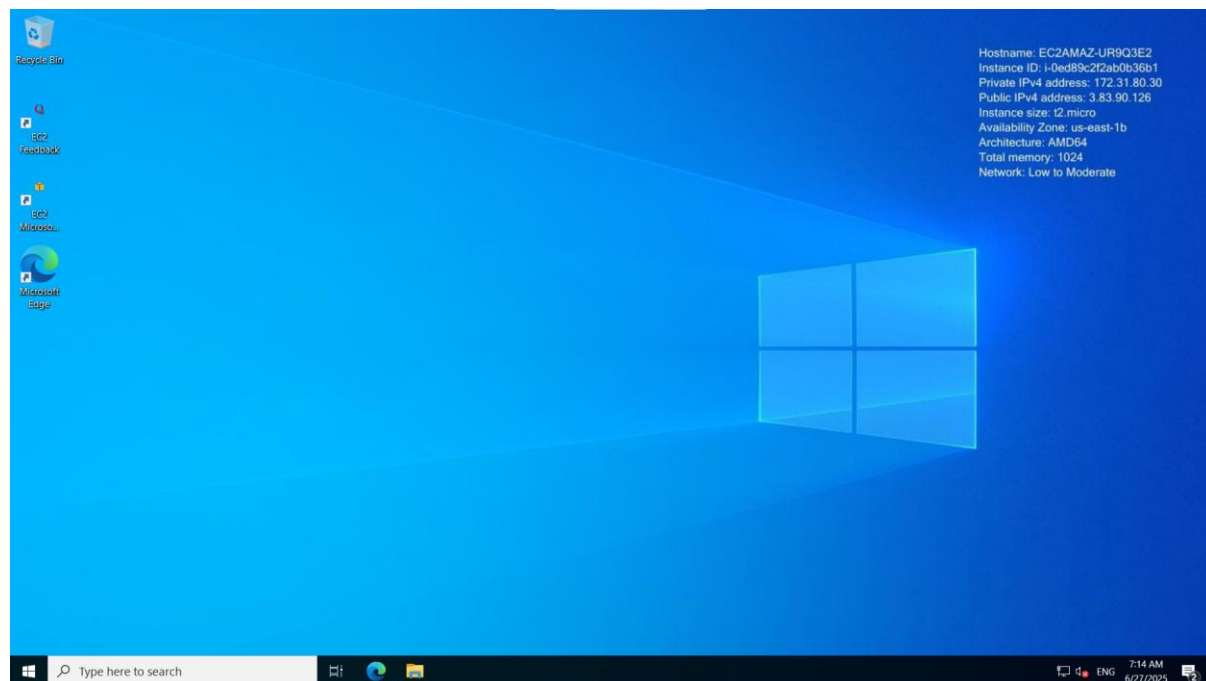
- Now search for the Remote Desktop Connection.
- In computer option paste your public IP address.
- Paste your username.
- Click on the connect.
- Then paste your password and click OK.



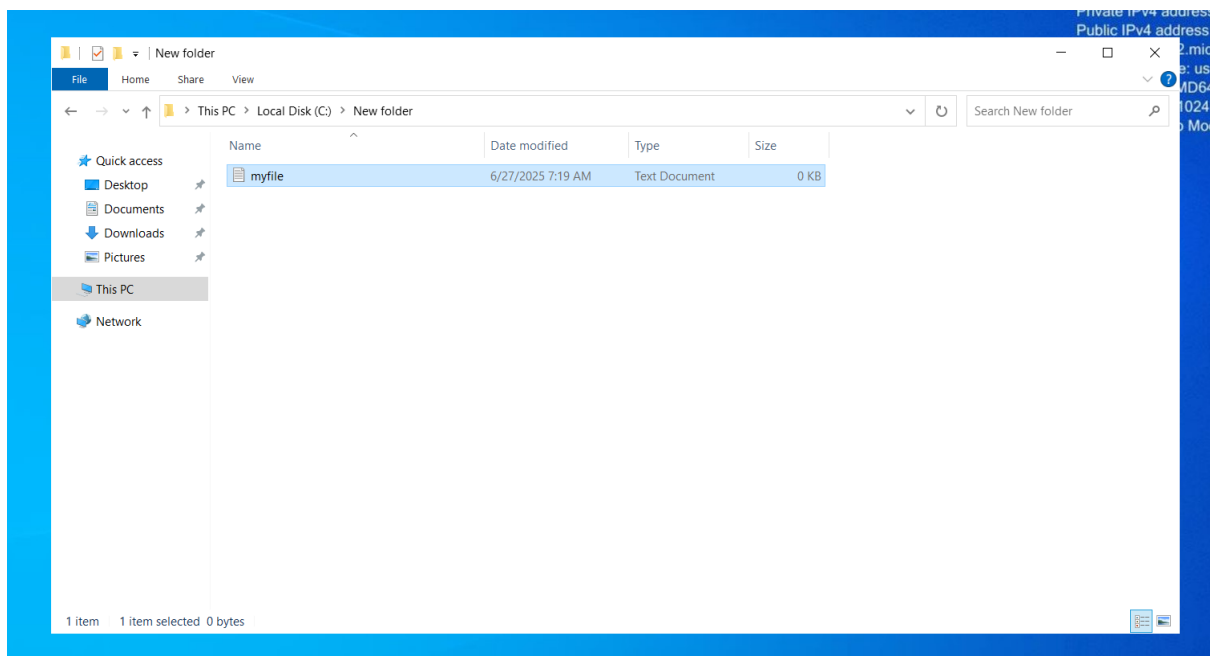
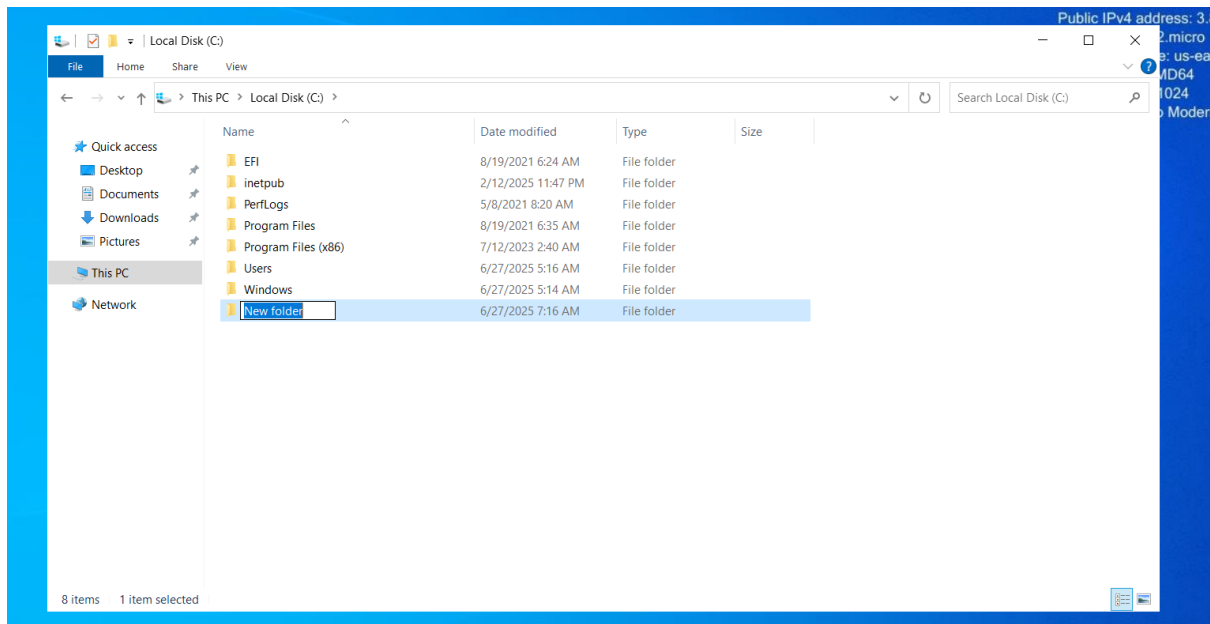
- Click OK.



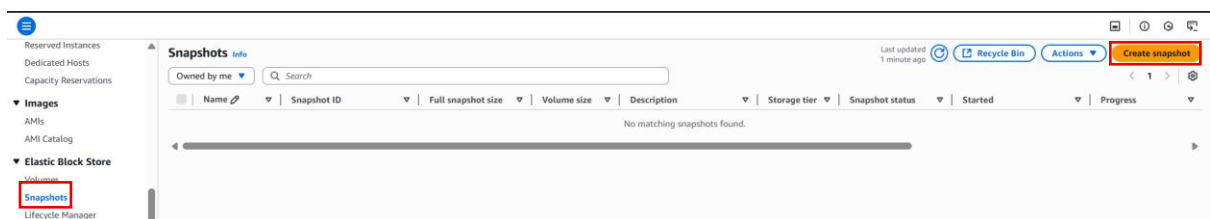
- Now we are successfully logged in to the virtual machine.



- Now here we will create a new folder in the root drive (C drive).
- And if you want to rename it you can do it.
- Now we will create a .txt file inside that folder.
- Put some content inside the file if you want to.



- Now let's go back to aws console.
- Go to EC2 > Snapshot.
- Create a snapshot.



- Here we can create snapshot from the 'volume and Instance' both.
- But here I am making using volume.
- Choose the volume in the volume ID filed.

- Give the description.
- Click Create snapshot.

Create snapshot Info

Create a point-in-time snapshot of an EBS volume and use it as a baseline for new volumes or for data backup. You can create snapshots from an individual volume, or you can create multi-volume snapshots from all of the volumes attached to an instance.

Source

Resource type Info

☒ **Volume**
Create a snapshot from a specific volume.

☐ **Instance**
Create multi-volume snapshots from an instance.

Volume ID
The volume from which to create the snapshot.

vol-06a239a07b23944c4
us-east-1b

Select a volume

vol-06a239a07b23944c4
us-east-1b

Add a description for your snapshot:

root-backup
255 characters maximum

Encryption Info

Not encrypted

Tags Info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add tag](#)

You can add 50 more tags.

[Cancel](#) [Create snapshot](#)

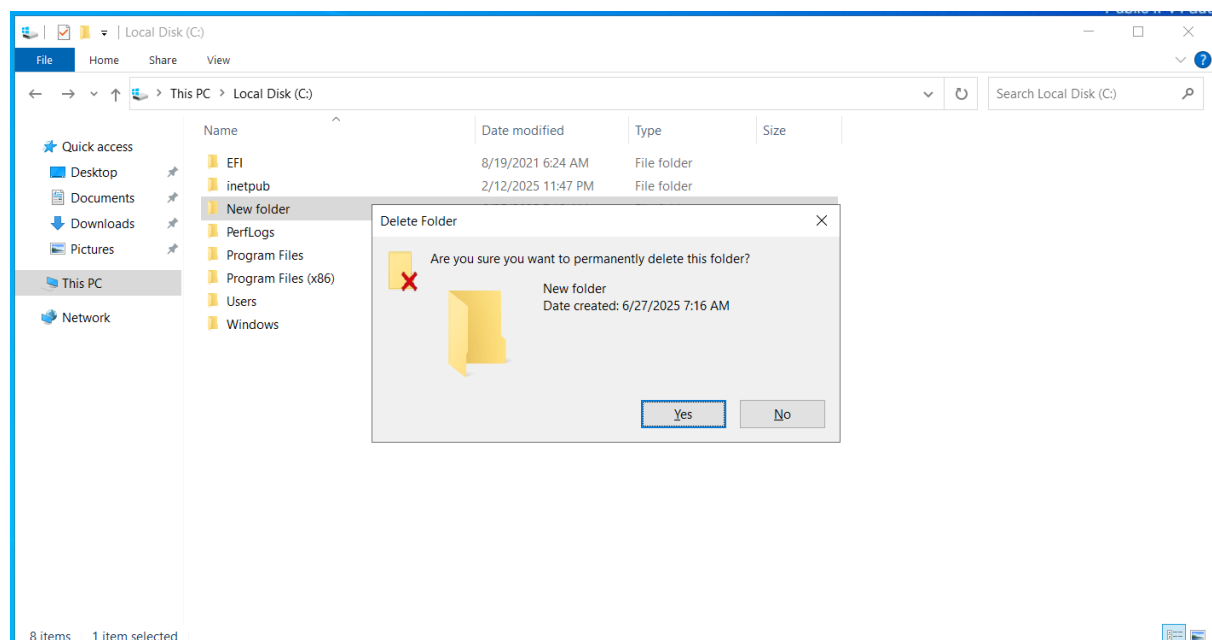
Successfully created snapshot snap-0bc6639163d149d09.

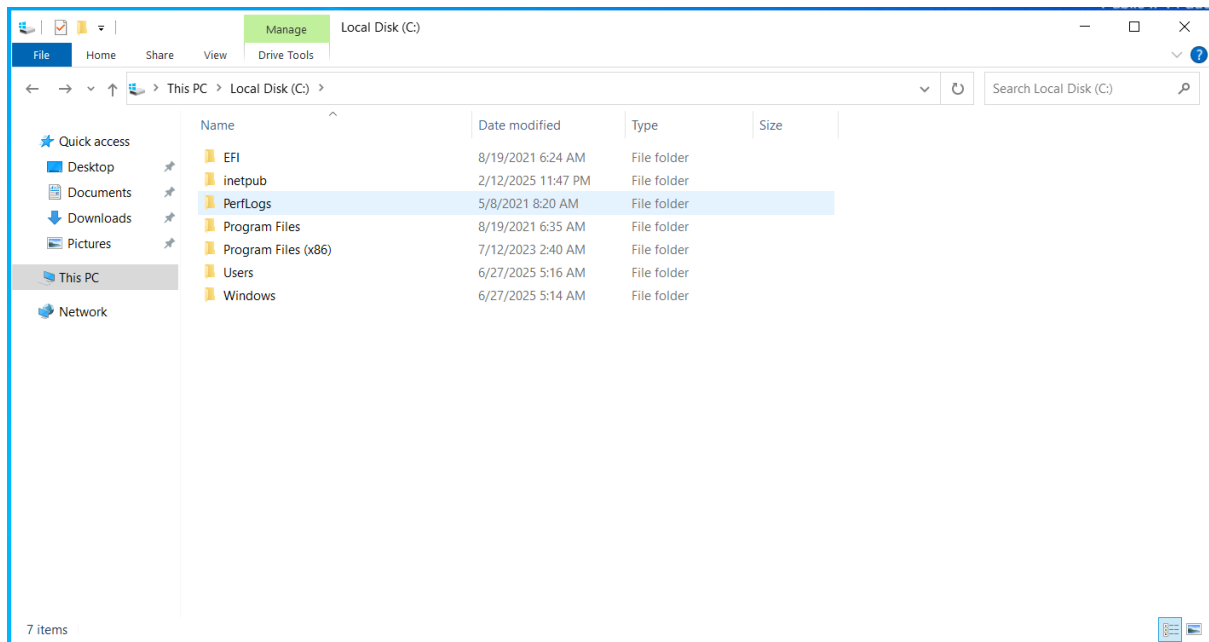
Less than a minute ago

[Recycle Bin](#) [Actions](#) [Create snapshot](#)

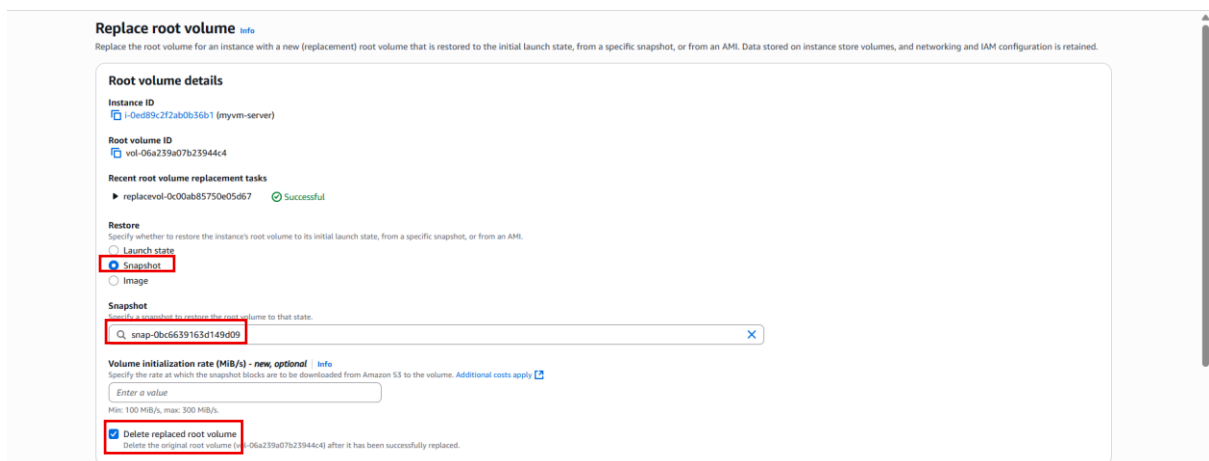
Owned by me	Search	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress
☐			snap-0bc6639163d149d09	-	30 GiB	root-backup	Standard	Pending	2025/06/27 13:17 GMT+5...	0%

- Now I will delete the folder permanently that we have created.

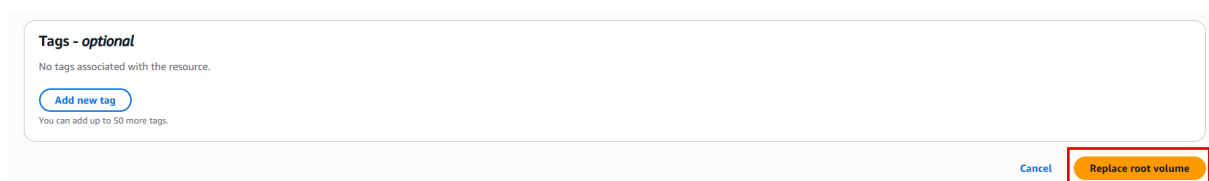




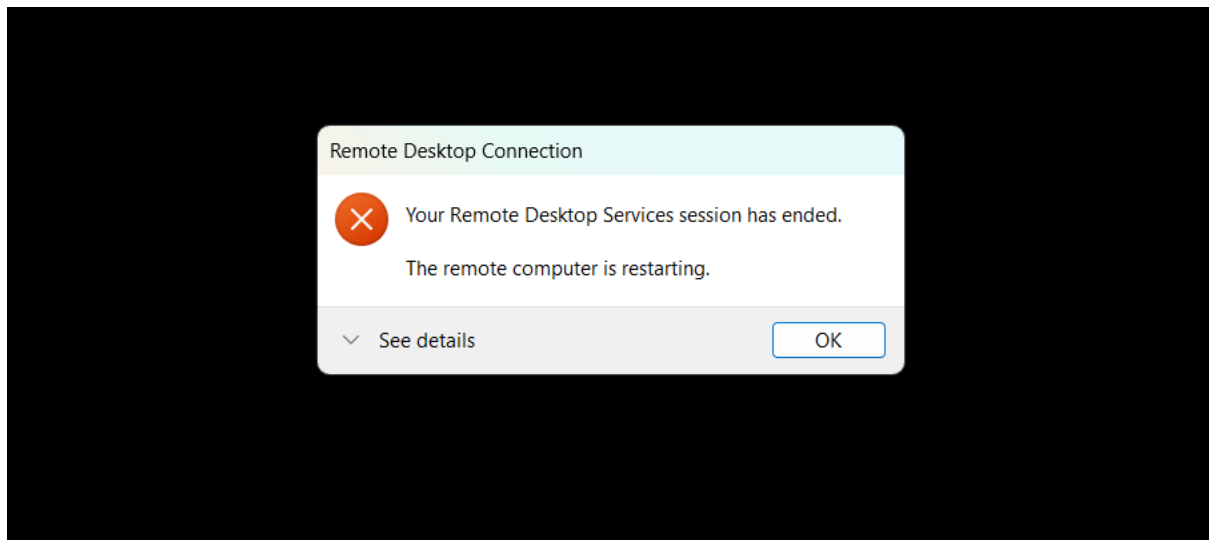
- Now go back to AWS console.
- Go to EC2 > Instance.
- Select the instance.
- Go to Action > Monitor and troubleshoot > Replace root volume.
- Choose the snapshot as an option.
- Choose the snapshot id that you have created.
- Tick the 'delete replaced root volume'.



- Click 'Replace root volume'.



- Your server will stop after you click replace root volume.



- Now log in again in the virtual machine.
- You can see we have successfully removed our folder.

